

Alexander Oh

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About Me

Software engineer passionate about building scalable, real-time systems for startups, government, and big tech. I thrive in fast-paced environments and enjoy using my creativity to innovate unique solutions. Excited to bring my expertise in payment systems, real-time data processing, and LLM fine-tuning to build impactful, high-performance products at scale.

Education

National University of Singapore, Singapore, SG

Aug 2023 – May 2027 (Expected)

Bachelor of Science in Computer Science (BS) | Cumulative GPA: 4.00 /5.00

- **Activities:** Raffles Hall: Tennis, Track and Frisbee IHG team

Professional Experience

Tetrix, United States, New York

Jan 2026 – Present

Backend Software Engineer

- Led development of a data extensibility framework supporting 15+ private market data models, enabling dynamic custom fields across fund portfolios, cap tables, and investor datasets.
- Designed a scalable custom field system that propagated across 6+ microservices, reducing schema change overhead by ~80% while maintaining consistency across analytics and frontend systems.
- Optimized time series computations by leveraging NumPy vectorization to process 100K+ rows of financial data, achieving ~5–10x performance improvements.
- Implemented custom formula support with DAG-based dependency resolution using topological sorting, eliminating circular dependencies and ensuring correct evaluation across complex financial metrics.
- Collaborated with the intelligent extraction team to process 50K+ documents, and built an Excel traceability system mapping LLM outputs to exact source coordinates with 100% accuracy, significantly improving auditability and trust.

TikTok, Singapore, SG

June 2025 – Nov 2025

Backend Software Engineer for User Growth (Paid Ads)

- Streamlined payment processes for ads using Go, Thrift/Kitex for RPC and Hertz for HTTP.
- Created an approval process for payments which incorporated >30 branches of conditional logic.
- Implemented a batch upload file system which applied a deduplication and error checking validation algorithm.
- The system handled >1000 files at a time and employed an asynchronous cronjob to maintain seamless user experience.

Asian Institute of Digital Finance (Fintech Startup), Singapore, SG

Jan 2025 – May 2025

Undergraduate Research Intern

- Completed a database migration to Bloomberg financial data for AIDF.
- Used Open AI api, fuzzy matching, pandas to match database variables.
- Wrote python scripts utilizing pandas to iteratively generate >3000 lines of SQL code for the new database.
- Updated and improved existing python backend to utilize new databases by rewriting old SQL queries and adding more features like error handling and multilingual input support.

Projects

Personal Website

Dec 2025 – Present

https://github.com/alexandtheoh/alexander_oh_website

- This project builds a personal portfolio website showcasing projects and experience, featuring a client-side LLM powered by WebLLM using a Gemma-2 2B model with a custom RAG pipeline.
- The RAG system is fully implemented in TypeScript with locally preprocessed embeddings, leveraging cosine similarity and top-k retrieval over ~10,000 indexed chunks to deliver fast, context-aware responses.

Barista Agent

Nov 2025 – Present

https://github.com/alexandtheoh/cafe_order_assistant

- Built an AI-powered barista agent that converts natural language orders into structured actions, handling inventory checks and payment processing across 20+ menu items.
- Implemented as a state-machine agent using OpenAI tool calls to manage multi-step interactions, achieving ~99% order accuracy.

Whatsapp LLM

May 2025 – Present

https://github.com/alexandtheoh/whatsapp_llm

- This project fine-tunes a large language model (Mistral 7b) to emulate a person using their whatsapp chat logs using the Unsloth library and QLoRA.
- The model was fine tuned using over 250,000 whatsapp messages with a custom training template in order for the model to emulate the sender's style and knowledge.

Skills and Interests

- **Languages:** Go, Javascript, Python, Java, C, VBA, Typescript, SQL
- **Frameworks:** Node.js, React.js, Pandas, Thrift, Hertz, Socket.io, Strapi, PM2, Cronjob
- **Tools:** Git, Docker, MQTT, RabbitMQ, Postman
- **Interests:** Tennis, Basketball, Golf, Hiking, Pokemon Showdown